3R Articulated Manipulator

Key Characteristics:

The manipulator under study is a three-revolute-joint (3R) serial robotic arm. All three joints are actuated revolute joints, making this a fully rotational, open-chain mechanism with three degrees of freedom. The system is designed for educational demonstration of robotic kinematics, dynamics, and point-to-point positioning.

Table 1: Key Characteristics of the 3R Manipulator

Parameter	Value / Description
Configuration	RRR — three revolute joints connected in series (serial chain)
Degrees of Freedom (DOF)	3
Joint 1 — θ_1	Rotation about vertical Z_0 axis (base rotation), range: 0° – 180°
Joint 2 — θ_2	Rotation about horizontal Z_1 axis (shoulder), range: 0° – 180°
Joint 3 — θ_3	Rotation about Z_2 axis (elbow), range: 0° – 180°
Base Height (d_1)	90 mm — vertical offset from ground to shoulder joint centre
Link 2 Length (l_2)	210 mm
Link 3 Length (l_3)	170 mm (includes end-effector structural body)
Maximum Vertical Reach	$d_1 + l_2 + l_3 = 90 + 210 + 170 = 470$ mm (arm fully extended upward, θ_2=90°, θ_3=0°)
Minimum Vertical Reach	$d_1 + l_2 - l_3 = 90 + 210 - 170 = 130$ mm (arm folded back, θ_2=90°, θ_3=180°)
End-Effector	Point at tip of Link 3 — no gripper; gripper profile is structural only
Actuators	MG995 Metal Gear Servo Motors × 3 — torque 9.4 kg·cm @ 6 V
Controller	ESP32 Microcontroller — PWM via ledc peripheral (14-bit resolution)
Link Material	Acrylic (PMMA) sheet — 5 mm thick, $\rho \approx 1180$ kg/m³
Link Cross-Section	Width (z-axis) = 60 mm, Height (y-axis) = 55 mm for Links 2 & 3
Link 1 Cross-Section	Width (z-axis) = 60 mm, Horizontal (x-axis) = 55 mm, Height (y-axis) = 90 mm
Design Software	SolidWorks (3D CAD and assembly)
Analysis Software	MATLAB (kinematics simulation, workspace, plots)
Firmware IDE	Arduino IDE (ESP32 PWM servo control)

DH Frame Assignment

Coordinate frames are assigned following the standard Denavit-Hartenberg convention. The Z-axis of each frame coincides with the joint rotation axis. The symbol ⊗ in the design sketch denotes a Z-axis directed into the page. Z_0 points vertically upward. Z_1 and Z_2 are horizontal and parallel (both pointing out of the plane in the side-view sketch), consistent with $\alpha_1 = 90°$ at joint 1 and $\alpha_2 = \alpha_3 = 0°$.

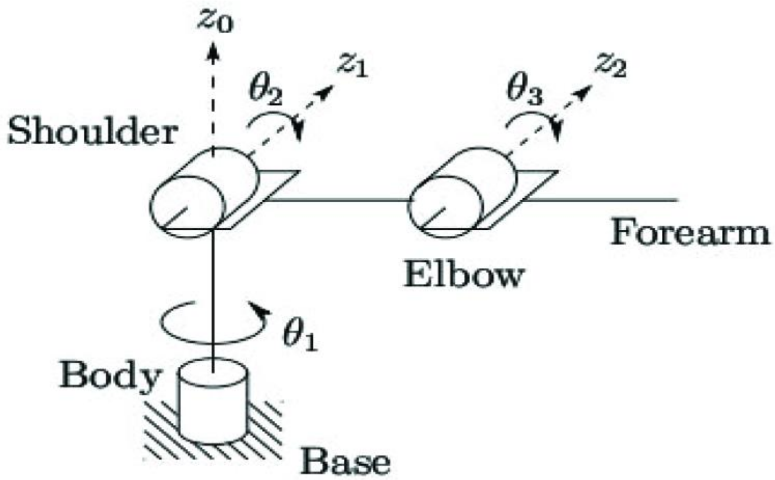

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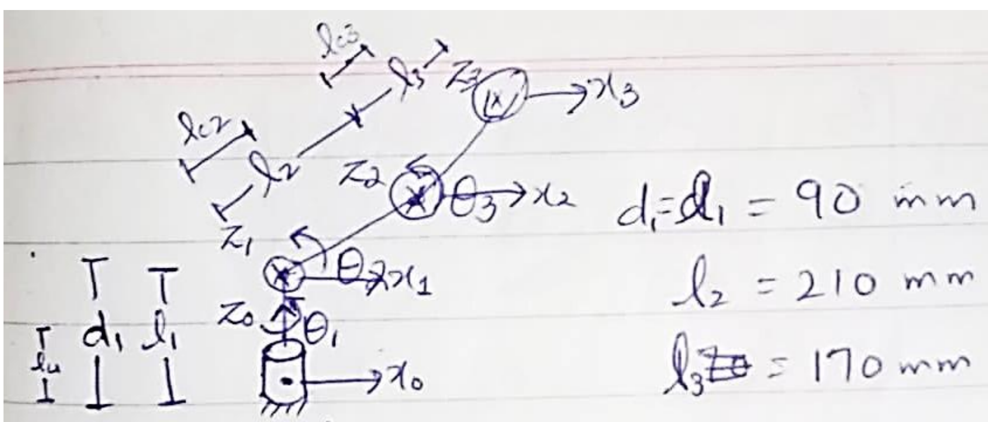
Figure 2: DH Frame Assignment Sketch



Home / Reference Configuration

Before executing any motion task, the manipulator is driven to a fixed reference pose known as the home configuration. This provides a consistent, safe starting state. The home joint angles are:

$$\theta_1 = 0°, \ \theta_2 = 0°, \ \theta_3 = 90°$$

Substituting into the forward kinematics (derived in Section 4):

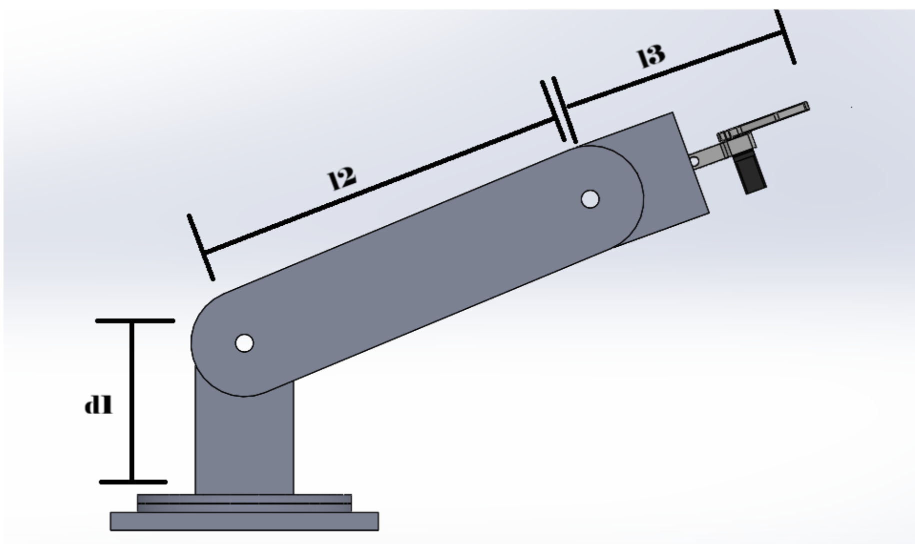
$$x = \cos 0° \cdot (210 \cos 0° + 170 \cos 90°) = 1 \cdot (210 + 0) = 210\, mm$$
$$y = \sin 0° \cdot (210 \cos 0° + 170 \cos 90°) = 0\, mm$$
$$z = 90 + 210 \sin 0° + 170 \sin 90° = 90 + 0 + 170 = 260\, mm$$

Home end-effector position: O_3 = (210, 0, 260) mm in the base frame.

Table 2: PWM Initialisation Values at Home Configuration

Joint	Home Angle	PWM Pulse Width
Joint 1 (θ_1)	0°	500 µs
Joint 2 (θ_2)	0°	500 µs
Joint 3 (θ_3)	90°	1500 µs

PWM frame frequency: 50 Hz. Range: 500 µs (0°) to 2500 µs (180°). On power-up, the ESP32 drives all channels to home values before accepting motion commands.



Homogeneous Transformation Matrices

A homogeneous transformation matrix T ∈ $\mathbb{R}^{4x4}$ encodes both the rotation and translation between two consecutive coordinate frames. Using the standard Denavit-Hartenberg convention, the transformation from frame i−1 to frame i is:

$$T_m = Rot(Z, \theta_i) \cdot Trans(Z, d_i) \cdot Trans(X, a_i) \cdot Rot(X, \alpha_i)$$

Expanding into 4×4 matrix form, where cθ = cosθ, sθ = sinθ, cα = cosα, sα = sinα:

$$T_i = \begin{bmatrix} c\theta_i & -s\theta_i\, c\alpha_i & s\theta_i\, s\alpha_i & a_i\, c\theta_i \\ s\theta_i & c\theta_i\, c\alpha_i & -c\theta_i\, s\alpha_i & a_i\, s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The shorthand notation $c_i = \cos\theta_i$, $s_i = \sin\theta_i$, $c_{23} = \cos(\theta_2+\theta_3)$, $s_{23} = \sin(\theta_2+\theta_3)$ is used throughout.

Individual Matrix ${}^0T^1$ (a_1=0, α_1=90°, d_1=90, θ_1)

Substituting $a_1=0$, $\alpha_1=90°$ ⟹ $\cos\alpha_1=0$, $\sin\alpha_1=1$, $d_1=90$ mm:

$${}^0T^1 = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & 90 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Individual Matrix ${}^1T^2$ (a_2=210, α_2=0°, d_2=0, θ_2)

Substituting $a_2=210$ mm, $\alpha_2=0°$ ⟹ $\cos\alpha_2=1$, $\sin\alpha_2=0$, $d_2=0$:

$${}^1T^2 = \begin{bmatrix} c_2 & -s_2 & 0 & 210c_2 \\ s_2 & c_2 & 0 & 210s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Individual Matrix ${}^2T^3$ (a_3=170, α_3=0°, d_3=0, θ_3)

Substituting $a_3=170$ mm, $\alpha_3=0°$, $d_3=0$:

$${}^2T^3 = \begin{bmatrix} c_3 & -s_3 & 0 & 170c_3 \\ s_3 & c_3 & 0 & 170s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Cumulative Matrix ${}^0T^2 = {}^0T^1 \cdot {}^1T^2$

Performing full row-by-column multiplication. Key element computations:

$$R(1,1) = c_1 \cdot c_2 + 0 \cdot s_2 + s_1 \cdot 0 = c_1 c_2$$

$$R(2,1) = s_1 \cdot c_2 + 0 \cdot s_2 + (-c_1) \cdot 0 = s_1 c_2$$
$$R(3,1) = 0 \cdot c_2 + 1 \cdot s_2 + 0 \cdot 0 = s_2$$
$$R(1,2) = c_1 \cdot (-s_2) + 0 \cdot c_2 + s_1 \cdot 0 = -c_1 s_2$$
$$R(1,3) = c_1 \cdot 0 + 0 \cdot 0 + s_1 \cdot 1 = s_1$$
$$R(2,3) = s_1 \cdot 0 + 0 \cdot 0 + (-c_1) \cdot 1 = -c_1$$
$$p\_x = c_1 \cdot 210 c_2 + 0 \cdot 210 s_2 + s_1 \cdot 0 + 0 = 210 c_1 c_2$$
$$p\_y = s_1 \cdot 210 c_2 + 0 \cdot 210 s_2 + (-c_1) \cdot 0 + 0 = 210 s_1 c_2$$
$$p\_z = 0 \cdot 210 c_2 + 1 \cdot 210 s_2 + 0 \cdot 0 + 90 = 210 s_2 + 90$$

Full result:

$$^0T^2 = \begin{bmatrix} c_1c_2 & -c_1s_2 & s_1 & 210c_1c_2 \\ s_1c_2 & -s_1s_2 & -c_1 & 210s_1c_2 \\ s_2 & c_2 & 0 & 210s_2 + 90 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Full End-Effector Transform $^0T^3 = {}^0T^2 \cdot {}^2T^3$

Using the angle-sum identities: $c_2c_3 - s_2s_3 = c_{23}$ and $s_2c_3 + c_2s_3 = s_{23}$. Key element derivations:

$$R(1,1) = c_1c_2 \cdot c_3 + (-c_1s_2) \cdot s_3 + s_1 \cdot 0 = c_1(c_2c_3 - s_2s_3) = c_1c_{23}$$
$$R(1,2) = c_1c_2 \cdot (-s_3) + (-c_1s_2) \cdot c_3 = -c_1(c_2s_3 + s_2c_3) = -c_1s_{23}$$
$$R(2,1) = s_1c_2c_3 - s_1s_2s_3 = s_1c_{23}$$
$$R(3,1) = s_2c_3 + c_2s_3 = s_{23}$$
$$p\_x = 170c_1c_{23} + 210c_1c_2 = c_1(210c_2 + 170c_{23})$$
$$p\_y = 170s_1c_{23} + 210s_1c_2 = s_1(210c_2 + 170c_{23})$$
$$p\_z = 170(s_2c_3 + c_2s_3) + 210s_2 + 90 = 90 + 210s_2 + 170s_{23}$$

Final result:

$$^0T^3 = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 & c_1(210c_2 + 170c_{23}) \\ s_1c_{23} & -s_1s_{23} & -c_1 & s_1(210c_2 + 170c_{23}) \\ s_{23} & c_{23} & 0 & 90 + 210s_2 + 170s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward Position Kinematics:

Forward kinematics maps a set of joint angles (θ_1, θ_2, θ_3) to the Cartesian position of the end-effector point O_3 in the base frame. The position is extracted from the fourth column of $^0T^3$.

DH Parameter Table

Table 3: Denavit-Hartenberg Parameters

Joint i	a_i (mm)	α_i (deg)	d_i (mm)	θ_i
1	0	90°	$d_1 = 90$	θ_1* (0° – 180°)
2	$l_2 = 210$	0°	0	θ_2* (0° – 180°)
3	$l_3 = 170$	0°	0	θ_3* (180° – 180°)

The asterisk (*) marks the variable (active) parameter. All other entries per row are constant geometric properties of the link.

Forward Kinematics Equations

Extracting the position column of ${}^0T^3$, the end-effector position O_3 = (x, y, z) in the base frame is:

$$x = c_1 \cdot (210c_2 + 170c_{23})$$
$$y = s_1 \cdot (210c_2 + 170c_{23})$$
$$z = 90 + 210s_2 + 170s_{23}$$

Physical interpretation: x and y share the same horizontal planar radius $r = 210c_2 + 170c_{23}$, scaled by $\cos\theta_1$ and $\sin\theta_1$ respectively. Joint 1 sweeps the arm through the horizontal plane; joints 2 and 3 control elevation.

Velocity Kinematics:

Velocity kinematics relates joint angular velocities to the end-effector linear and angular velocities through the Jacobian matrix. The instructor's explicit geometric Jacobian method is followed throughout, using skew-symmetric matrix notation S(Z) in place of direct cross products.

Jacobian Method Overview

For each revolute joint i, the Jacobian column is:

$$J_{vi} = S(Z_{i-1}) \cdot (O_3 - O_{i-1}) \quad [linear\ velocity\ contribution]$$
$$J_{\omega i} = Z_{i-1} \quad [angular\ velocity\ contribution]$$

where S(Z) is the skew-symmetric matrix of $Z = [z\_x, z\_y, z\_z]^T$:

$$S(Z) = \begin{bmatrix} 0 & -z\_z & z\_y \\ z\_z & 0 & -z\_x \\ -z\_y & z\_x & 0 \end{bmatrix}$$

The end-effector position is denoted O_3 (not P). All subtractions $O_3 - O_i$ are performed term-by-term without premature cancellation.

Step 1: Z-Axis Vectors (from ^0T_i rotation column 3)

$$Z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$Z_1 = Third\ column\ of\ R\ in\ {}^0T^1 = \begin{bmatrix} s_1 \\ -c_1 \\ 0 \end{bmatrix}$$

$$Z_2 = Third\ column\ of\ R\ in\ {}^0T^2 = \begin{bmatrix} s_1 \\ -c_1 \\ 0 \end{bmatrix}$$

Note: $Z_1 = Z_2$ because $\alpha_2 = 0°$ introduces no twist between frames 1 and 2.

Step 2: Origin Vectors (from ^0T_i position column 4)

$$O_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$O_1 = \begin{bmatrix} 0 \\ 0 \\ 90 \end{bmatrix}$$

$$O_2 = \begin{bmatrix} 210c_1c_2 \\ 210s_1c_2 \\ 90 + 210s_2 \end{bmatrix}$$

$$O_3 = \begin{bmatrix} c_1(210c_2 + 170c_{23}) \\ s_1(210c_2 + 170c_{23}) \\ 90 + 210s_2 + 170s_{23} \end{bmatrix}$$

Step 3: Skew-Symmetric Matrices S(Z)

$S(Z_0)$: $Z_0 = [0, 0, 1]^T \Longrightarrow$ z_x=0, z_y=0, z_z=1

$$S(Z_0) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

S(Z_1): $Z_1 = [s_1, -c_1, 0]^T \Longrightarrow$ z_x=s_1, z_y=$-c_1$, z_z=0

$$S(Z_1) \;=\; \begin{bmatrix} 0 & 0 & -c_1 \\ 0 & 0 & -s_1 \\ c_1 & s_1 & 0 \end{bmatrix}$$

S(Z_2) = S(Z_1) (same vector)

$$S(Z_2) \;=\; \begin{bmatrix} 0 & 0 & -c_1 \\ 0 & 0 & -s_1 \\ c_1 & s_1 & 0 \end{bmatrix}$$

$O_3 - O_i$ Subtractions (term by term, no premature cancellation)

Define shorthand: $P = 210c_2 + 170c_{23}$ and $Q = 210s_2 + 170s_{23}$.

$O_3 - O_0$

$$Row\ 1\text{: } c_1P \;-\; 0 \;=\; c_1P$$

$$Row\ 2\text{: } s_1P \;-\; 0 \;=\; s_1P$$

$$Row\ 3\text{: } (90 + Q) \;-\; 0 \;=\; 90 + Q$$

$$O_3 \;-\; O_0 \;=\; \begin{bmatrix} c_1P \\ s_1P \\ 90 \;+\; Q \end{bmatrix}$$

$O_3 - O_1$

$$Row\ 1\text{: } c_1P \;-\; 0 \;=\; c_1P$$

$$Row\ 2\text{: } s_1P \;-\; 0 \;=\; s_1P$$

$$Row\ 3\text{: } (90 + Q) \;-\; 90 \;=\; Q$$

$$O_3 \;-\; O_1 \;=\; \begin{bmatrix} c_1P \\ s_1P \\ Q \end{bmatrix}$$

$O_3 - O_2$

$$Row\ 1\text{: } c_1(210c_2 + 170c_{23}) \;-\; 210c_1c_2 \;=\; 210c_1c_2 \;+\; 170c_1c_{23} \;-\; 210c_1c_2$$
$$=\; (210c_1c_2 - 210c_1c_2) \;+\; 170c_1c_{23} \;=\; 170c_1c_{23}$$

$$Row\ 2\text{: } s_1(210c_2 + 170c_{23}) \;-\; 210s_1c_2 \;=\; (210s_1c_2 - 210s_1c_2) \;+\; 170s_1c_{23}$$
$$=\; 170s_1c_{23}$$

$$Row\ 3\text{: } (90 + Q) \;-\; (90 + 210s_2) \;=\; (90 - 90) \;+\; (210s_2 + 170s_{23} - 210s_2)$$
$$=\; (210s_2 - 210s_2) \;+\; 170s_{23} \;=\; 170s_{23}$$

$$O_3 - O_2 = \begin{bmatrix} 170c_1c_{23} \\ 170s_1c_{23} \\ 170s_{23} \end{bmatrix}$$

Step 5: Linear Jacobian Columns $J_{vi} = S(Z_{i-1})(O_3 - O_{i-1})$

Column 1: $J_{v1} = S(Z_0)\cdot(O_3 - O_0)$

Full row-by-row multiplication:

$$Row\ 1\text{: } (0)(c_1P) + (-1)(s_1P) + (0)(90+Q) = -s_1P$$
$$Row\ 2\text{: } (1)(c_1P) + (0)(s_1P) + (0)(90+Q) = c_1P$$
$$Row\ 3\text{: } (0)(c_1P) + (0)(s_1P) + (0)(90+Q) = 0$$

$$J_{v1} = \begin{bmatrix} -s_1P \\ c_1P \\ 0 \end{bmatrix}$$

Column 2: $J_{v2} = S(Z_1)\cdot(O_3 - O_1)$

$$Row\ 1\text{: } (0)(c_1P) + (0)(s_1P) + (-c_1)(Q) = -c_1Q$$
$$Row\ 2\text{: } (0)(c_1P) + (0)(s_1P) + (-s_1)(Q) = -s_1Q$$
$$Row\ 3\text{: } (c_1)(c_1P) + (s_1)(s_1P) + (0)(Q) = {c_1}^2P + {s_1}^2P = P({c_1}^2 + {s_1}^2) = P$$

$$J_{v2} = \begin{bmatrix} -c_1Q \\ -s_1Q \\ P \end{bmatrix}$$

Column 3: $J_{v3} = S(Z_2)\cdot(O_3 - O_2)$

$$Row\ 1\text{: } (0)(170c_1c_{23}) + (0)(170s_1c_{23}) + (-c_1)(170s_{23}) = -170c_1s_{23}$$
$$Row\ 2\text{: } (0)(170c_1c_{23}) + (0)(170s_1c_{23}) + (-s_1)(170s_{23}) = -170s_1s_{23}$$
$$Row\ 3\text{: } (c_1)(170c_1c_{23}) + (s_1)(170s_1c_{23}) + (0)(170s_{23}) = 170c_{23}({c_1}^2 + {s_1}^2)$$
$$= 170c_{23}$$

$$J_{v3} = \begin{bmatrix} -170c_1s_{23} \\ -170s_1s_{23} \\ 170c_{23} \end{bmatrix}$$

Step 6: Angular Jacobian Columns $J\omega_i = Z_{i-1}$

$$J\omega_1 = Z_0 = [0,\ 0,\ 1]^{\mathrm{T}}$$
$$J\omega_2 = Z_1 = [s_1, -c_1, 0]^{\mathrm{T}}$$
$$J\omega_3 = Z_2 = [s_1, -c_1, 0]^{\mathrm{T}}$$

Step 7: Full Geometric Jacobian J (6×3)

Assembling all columns, where $P = 210c_2+170c_{23}$ and $Q = 210s_2+170s_{23}$:

$$J = \begin{bmatrix} -s_1P & -c_1Q & -170c_1s_{23} & \leftarrow v\_x \\ c_1P & -s_1Q & -170s_1s_{23} & \leftarrow v\_y \\ 0 & P & 170c_{23} & \leftarrow v\_z \\ 0 & s_1 & s_1 & \leftarrow \omega\_x \\ 0 & -c_1 & -c_1 & \leftarrow \omega\_y \\ 1 & 0 & 0 & \leftarrow \omega\_z \end{bmatrix}$$

Analytical (Linear Velocity) Jacobian J_v (3×3)

The upper 3×3 sub-matrix gives the linear velocity mapping $\dot{x} = J_v \dot{\theta}$:

$$J_v = \begin{bmatrix} -s_1P & -c_1Q & -170c_1s_{23} \\ c_1P & -s_1Q & -170s_1s_{23} \\ 0 & P & 170c_{23} \end{bmatrix}$$

Singularity Analysis

A singularity occurs when $\det(J_v) = 0$. The arm cannot generate velocity in all directions at such configurations.

Determinant Calculation — Cofactor Expansion Along Row 3

$$det(J_v) = 0 \cdot C_{31} + P \cdot C_{32} + 170c_{23} \cdot C_{33}$$

Cofactor C_{32}:

$$\begin{aligned} C_{32} &= (-1)^5\, det[[-s_1P, -170c_1s_{23}], [c_1P, -170s_1s_{23}]] \\ &= -1 \cdot [(-s_1P)(-170s_1s_{23}) - (-170c_1s_{23})(c_1P)] \\ &= -1 \cdot [170s_1{}^2Ps_{23} + 170c_1{}^2Ps_{23}] \\ &= -1 \cdot 170Ps_{23}(s_1{}^2 + c_1{}^2) = -170Ps_{23} \end{aligned}$$

Cofactor C_{33}:

$$\begin{aligned} C_{33} &= (+1)\, det[[-s_1P, -c_1Q], [c_1P, -s_1Q]] \\ &= (-s_1P)(-s_1Q) - (-c_1Q)(c_1P) \\ &= s_1{}^2PQ + c_1{}^2PQ = PQ(s_1{}^2 + c_1{}^2) = PQ \end{aligned}$$

Substituting:

$$\begin{aligned} det(J_v) &= P \cdot (-170Ps_{23}) + 170c_{23} \cdot PQ \\ &= 170P(-Ps_{23} + Qc_{23}) \end{aligned}$$

$$= 170P[c_{23}(210s_2 + 170s_{23}) - s_{23}(210c_2 + 170c_{23})]$$
$$= 170P \cdot 210[c_{23}s_2 - s_{23}c_2]$$
$$= 170 \cdot 210 \cdot P \cdot sin(\theta_2 - (\theta_2 + \theta_3))$$
$$= -35700 \cdot P \cdot sin\theta_3$$

$$det(J_v) = -35700 \cdot (210c_2 + 170c_{23}) \cdot sin\theta_3$$

Singular Configurations:

Singularity 1 — Elbow Singularity:

$$sin\theta_3 = 0 \quad \Rightarrow \quad \theta_3 = 0° \ or \ \theta_3 = 180°$$

At θ_3=0° the arm is fully extended; at θ_3=180° it is fully folded. Joints 2 and 3 become collinear and lose one independent control direction in the (r,z) plane.

Singularity 2 — Workspace Boundary:

$$P = 0 \quad \Rightarrow \quad 210c_2 + 170c_{23} = 0$$

The end-effector lies on the vertical Z_0 axis (r = 0). The azimuthal angle θ_1 is undefined and joint 1 becomes singular. Any θ_1 can produce the same x=y=0 position.

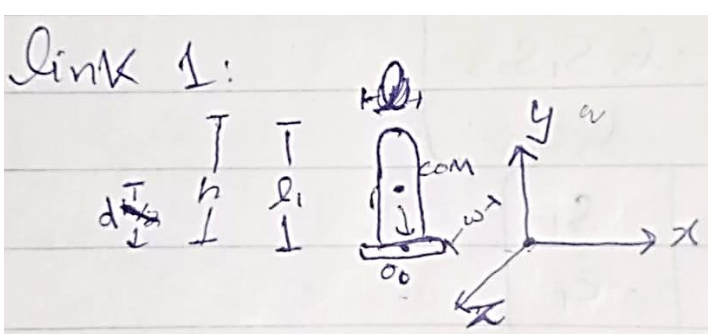
Dynamics:

The equation of motion of the 3R manipulator, derived from the Euler-Lagrange formulation, is:

$$M(q)\ddot{a}q + B(q)[\dot{q}q] + C(q)[q^2] + G(q) = \tau$$

where q = $[\theta_1,\theta_2,\theta_3]^T$ is the joint vector, M(q) is the 3×3 symmetric mass matrix, B(q) is the Coriolis matrix, C(q) is the centrifugal matrix, G(q) is the gravity vector, and τ is the joint torque vector.

Inertia Tensor — Link 1



Link 1 is the vertical column. Its dimensions in the Cartesian frame are:

- a_1 = 55 mm (along x, horizontal)
- b_1 = 90 mm (along y, vertical — this is the link length)
- c_1 = 60 mm (along z, out of plane)

The inertia tensor at the centre of mass (COM), derived by direct triple integration over limits $-a_1/2$ to $a_1/2$ in x, $-b_1/2$ to $b_1/2$ in y, $-c_1/2$ to $c_1/2$ in z:

General derivation of I_xx (same procedure for other components):

$$I_{xx} \;=\; \rho \int\int\int\; (y^2 + z^2)\, dV$$
$$=\; \rho \int_{-}{}^{1}/{}^{2a} \int_{-}{}^{b}/{}^{2b}/{}^{2} \int_{-}{}^{c}/{}^{2c}/{}^{2}\, (y^2 + z^2)\, dz\, dy\, dx$$
$$=\; \rho \cdot a \cdot \int_{-}{}^{b}/{}^{2b}/{}^{2}\, [y^2 z \;+\; z^3/3]_{-}{}^{c}/{}^{2c}/{}^{2}\, dy$$
$$=\; \rho \cdot a \cdot \int_{-}{}^{b}/{}^{2b}/{}^{2}\, (y^2 c \;+\; c^3/12)\, dy$$
$$=\; \rho \cdot a \cdot [cy^3/3 \;+\; c^3 y/12]_{-}{}^{b}/{}^{2b}/{}^{2}$$
$$=\; \rho \cdot a \cdot (cb^3/12 \;+\; c^3 b/12) \;=\; \rho \cdot abc \cdot (b^2 + c^2)/12 \;=\; m(b^2 + c^2)/12$$

Applying to Link 1 (a_1=55, b_1=90, c_1=60, mass m_1):

$$I_1xx\_com \;=\; m_1({b_1}^2 + {c_1}^2)/12 \;=\; m_1(90^2 + 60^2)/12 \;=\; m_1(8100 + 3600)/12$$
$$=\; 11700 m_1/12 \;=\; 975 m_1\, mm^2$$
$$I_1yy\_com \;=\; m_1({a_1}^2 + {c_1}^2)/12 \;=\; m_1(55^2 + 60^2)/12 \;=\; m_1(3025 + 3600)/12$$
$$=\; 6625 m_1/12 \;=\; 552.08 m_1\, mm^2$$
$$I_1zz\_com \;=\; m_1({a_1}^2 + {b_1}^2)/12 \;=\; m_1(55^2 + 90^2)/12 \;=\; m_1(3025 + 8100)/12$$
$$=\; 11125 m_1/12 \;=\; 927.08 m_1\, mm^2$$

$$I_1\_com \;=\; \begin{bmatrix} 975m_1 & 0 & 0 \\ 0 & 552.08m_1 & 0 \\ 0 & 0 & 927.08m_1 \end{bmatrix}$$

Parallel-Axis Theorem — shift from COM to bottom face (motor mount). The COM shifts downward along y by $b_1/2$ = 45 mm, so d = $[0, 45, 0]^T$:

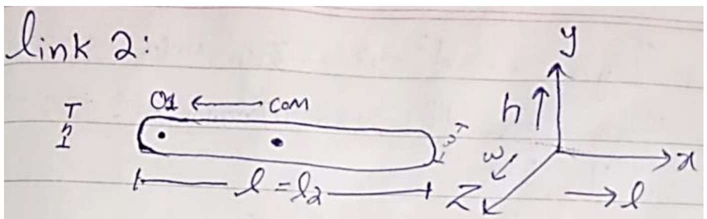
$$I_1xx \;=\; I_1xx\_com \;+\; m_1 \cdot (45^2) \;=\; 975m_1 \;+\; 2025m_1 \;=\; 3000m_1\, mm^2$$
$$I_1yy \;=\; I_1yy\_com \;+\; 0 \;=\; 552.08m_1\, mm^2 \qquad (shift\ is\ along\ y \;\Longrightarrow\; Iyy\ unchanged)$$
$$I_1zz \;=\; I_1zz\_com \;+\; m_1 \cdot (45^2) \;=\; 927.08m_1 \;+\; 2025m_1 \;=\; 2952.08m_1\, mm^2$$

$$I_1 = \begin{bmatrix} 3000m_1 & 0 & 0 \\ 0 & 552.08m_1 & 0 \\ 0 & 0 & 2952.08m_1 \end{bmatrix}$$

Scalar effective inertia for dynamics: Joint 1 rotates about the y-axis, so the rotation axis normal is $n_1 = [0,1,0]^T$:

$$I_1 = n_1^{\mathrm{T}} \cdot I_1 \cdot n_1 = I_1yy\_com = 552.08\, m_1\ mm^2$$

Inertia Tensor — Link 2



Link 2 is horizontal. Dimensions: a_2=210 mm (x, length), b_2=55 mm (y), c_2=60 mm (z).

$$I_2xx\_com = m_2(b_2{}^2 + c_2{}^2)/12 = m_2(55^2 + 60^2)/12 = m_2(3025 + 3600)/12 = 552.08m_2\, mm^2$$

$$I_2yy\_com = m_2(a_2{}^2 + c_2{}^2)/12 = m_2(210^2 + 60^2)/12 = m_2(44100 + 3600)/12 = 47700m_2/12 = 3975m_2\, mm^2$$

$$I_2zz\_com = m_2(a_2{}^2 + b_2{}^2)/12 = m_2(210^2 + 55^2)/12 = m_2(44100 + 3025)/12 = 47125m_2/12 = 3927.08m_2\, mm^2$$

PAT — shift from COM to left end (Joint 2 mount) along x by $a_2/2$ = 105 mm. $d = [105,0,0]^T$:

$$I_2xx = I_2xx\_com = 552.08m_2\, mm^2 \qquad (shift\ along\ x \implies Ixx\ unchanged)$$

$$I_2yy = I_2yy\_com + m_2 \cdot (105^2) = 3975m_2 + 11025m_2 = 15000m_2\, mm^2$$

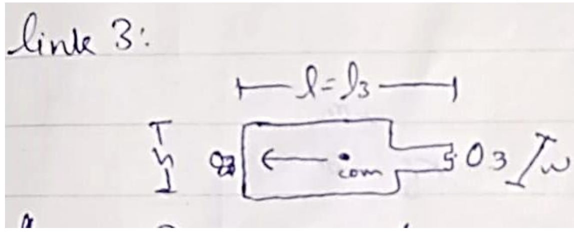
$$I_2zz = I_2zz\_com + m_2 \cdot (105^2) = 3927.08m_2 + 11025m_2 = 14952.08m_2\, mm^2$$

$$I_2 = \begin{bmatrix} 552.08m_2 & 0 & 0 \\ 0 & 15000m_2 & 0 \\ 0 & 0 & 14952.08m_2 \end{bmatrix}$$

Scalar effective inertia: Joint 2 rotates about the z-axis, $n_2 = [0,0,1]^T$:

$$I_2 = n_2^{\mathrm{T}} \cdot I_2 \cdot n_2 = I_2zz = 14952.08\, m_2\ mm^2$$

Inertia Tensor — Link 3



Link 3 is horizontal. Dimensions: a_3=170 mm (x), b_3=55 mm (y), c_3=60 mm (z).

$$I_3xx\_com = m_3(55^2 + 60^2)/12 = 552.08m_3\, mm^2$$
$$I_3yy\_com = m_3(170^2 + 60^2)/12 = m_3(28900 + 3600)/12 = 32500m_3/12 = 2708.33m_3\, mm^2$$
$$I_3zz\_com = m_3(170^2 + 55^2)/12 = m_3(28900 + 3025)/12 = 31925m_3/12 = 2660.42m_3\, mm^2$$

PAT — shift from COM to left end (Joint 3 mount) along x by $a_3/2$ = 85 mm. d = $[85,0,0]^T$:

$$I_3xx = 552.08m_3\, mm^2 \qquad (shift\ along\ x \implies Ixx\ unchanged)$$
$$I_3yy = 2708.33m_3 + m_3 \cdot (85^2) = 2708.33m_3 + 7225m_3 = 9933.33m_3\, mm^2$$
$$I_3zz = 2660.42m_3 + m_3 \cdot (85^2) = 2660.42m_3 + 7225m_3 = 9885.42m_3\, mm^2$$

$$I_3 = \begin{bmatrix} 552.08m_3 & 0 & 0 \\ 0 & 9933.33m_3 & 0 \\ 0 & 0 & 9885.42m_3 \end{bmatrix}$$

Scalar effective inertia: Joint 3 also rotates about z-axis, $n_3 = [0,0,1]^T$:

$$I_3 = I_3zz = 9885.42\, m_3\ mm^2$$

Table 4: Summary of Inertia Tensor Values at Motor Mount Points

Link	I_xx (mm²)	I_yy (mm²)	I_zz (mm²)	Scalar I_i used in dynamics
Link 1	3000 m_1	552.08 m_1 (=I_1)	2952.08 m_1	I_1 = 552.08 m_1 (rot. about y)
Link 2	552.08 m_2	15000 m_2	14952.08 m_2 (=I_2)	I_2 = 14952.08 m_2 (rot. about z)
Link 3	552.08 m_3	9933.33 m_3	9885.42 m_3 (=I_3)	I_3 = 9885.42 m_3 (rot. about z)

Centre of Mass Positions

COM half-lengths: $lc_1 = b_1/2$ = 45 mm, $lc_2 = a_2/2$ = 105 mm, $lc_3 = a_3/2$ = 85 mm.

COM positions in the DH base frame:

$$C_1 = \begin{bmatrix} 0 \\ 0 \\ lc_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 45 \end{bmatrix}$$

(C_1 lies on the Z_0 axis — joint 1 rotation produces no linear velocity at this COM)

$$C_2 = \begin{bmatrix} lc_2\, c_1 c_2 \\ lc_2\, s_1 c_2 \\ d_1 + lc_2\, s_2 \end{bmatrix} = \begin{bmatrix} 105 c_1 c_2 \\ 105 s_1 c_2 \\ 90 + 105 s_2 \end{bmatrix}$$

Shorthand: $Pc = l_2c_2 + lc_3c_{23} = 210c_2 + 85c_{23}$, $Qc = l_2s_2 + lc_3s_{23} = 210s_2 + 85s_{23}$

$$C_3 = \begin{bmatrix} c_1 \cdot Pc \\ s_1 \cdot Pc \\ 90 + Qc \end{bmatrix} = \begin{bmatrix} c_1(210c_2 + 85c_{23}) \\ s_1(210c_2 + 85c_{23}) \\ 90 + 210s_2 + 85s_{23} \end{bmatrix}$$

COM Jacobians

The COM Jacobian for link i uses the same S(Z) method but targets C_i instead of O_3. Columns after joint i are zero (joints beyond i do not move COM of link i).

JvC_1 and $J\omega C_1$ (Link 1 — column 1 only, rest zero)

Column 1: $Jv_{1}\_C_1 = S(Z_0)(C_1 - O_0)$

$$C_1 - O_0 = [0, 0, 45]^{\mathrm{T}} - [0, 0, 0]^{\mathrm{T}} = [0, 0, 45]^{\mathrm{T}}$$

$$Row\ 1\text{: } (0)(0) + (-1)(0) + (0)(45) = 0$$

$$Row\ 2\text{: } (1)(0) + (0)(0) + (0)(45) = 0$$

$$Row\ 3\text{: } (0)(0) + (0)(0) + (0)(45) = 0$$

C_1 lies on the Z_0 axis $\Longrightarrow$ joint 1 rotation produces zero linear velocity at C_1.

$$JvC_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad J\omega C_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

JvC_2 and $J\omega C_2$ (Link 2 — columns 1 and 2; column 3 = 0)

Column 1: $Jv_{1}\_C_2 = S(Z_0)(C_2 - O_0)$

$$C_2 - O_0 = [105c_1c_2, 105s_1c_2, 90 + 105s_2]^{\mathrm{T}}$$

$$Row\ 1\text{: } (0)(105c_1c_2) + (-1)(105s_1c_2) + (0)(90 + 105s_2) = -105s_1c_2$$

$$Row\ 2\text{: } (1)(105c_1c_2) + (0)(105s_1c_2) + (0)(90 + 105s_2) = 105c_1c_2$$

$$Row\ 3\text{: } (0)(105c_1c_2) + (0)(105s_1c_2) + (0)(90 + 105s_2) \ = \ 0$$

Column 2: $Jv_2\_C_2 = S(Z_1)(C_2 - O_1)$

$$C_2 - O_1 \ = \ [105c_1c_2 - 0, 105s_1c_2 - 0, (90 + 105s_2) - 90]^{\mathrm{T}}$$
$$= \ [105c_1c_2, 105s_1c_2, 105s_2]^{\mathrm{T}}$$
$$Row\ 1\text{: } (0)(105c_1c_2) + (0)(105s_1c_2) + (-c_1)(105s_2) \ = \ -105c_1s_2$$
$$Row\ 2\text{: } (0)(105c_1c_2) + (0)(105s_1c_2) + (-s_1)(105s_2) \ = \ -105s_1s_2$$
$$Row\ 3\text{: } (c_1)(105c_1c_2) + (s_1)(105s_1c_2) + (0)(105s_2) \ = \ 105c_2({c_1}^2 + {s_1}^2) \ = \ 105c_2$$

$$JvC_2 \ = \ \begin{bmatrix} -105s_1c_2 & -105c_1s_2 & 0 \\ 105c_1c_2 & -105s_1s_2 & 0 \\ 0 & 105c_2 & 0 \end{bmatrix}$$

$$J\omega C_2 \ = \ \begin{bmatrix} 0 & s_1 & 0 \\ 0 & -c_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

JvC_3 and $J\omega C_3$ (Link 3 — all three columns)

Column 1: $Jv_1\_C_3 = S(Z_0)(C_3 - O_0)$

$$C_3 - O_0 \ = \ [c_1Pc, s_1Pc, 90 + Qc]^{\mathrm{T}}$$
$$Row\ 1\text{: } (-1)(s_1Pc) \ = \ -s_1Pc$$
$$Row\ 2\text{: } (1)(c_1Pc) \ = \ c_1Pc$$
$$Row\ 3\text{: } 0$$

Column 2: $Jv_2\_C_3 = S(Z_1)(C_3 - O_1)$

$$C_3 - O_1 \ = \ [c_1Pc, s_1Pc, Qc]^{\mathrm{T}}$$
$$Row\ 1\text{: } (0)(c_1Pc) + (0)(s_1Pc) + (-c_1)(Qc) \ = \ -c_1Qc$$
$$Row\ 2\text{: } (0)(c_1Pc) + (0)(s_1Pc) + (-s_1)(Qc) \ = \ -s_1Qc$$
$$Row\ 3\text{: } (c_1)(c_1Pc) + (s_1)(s_1Pc) + (0)(Qc) \ = \ Pc({c_1}^2 + {s_1}^2) \ = \ Pc$$

Column 3: $Jv_3\_C_3 = S(Z_2)(C_3 - O_2)$

$$C_3 - O_2 \ = \ [85c_1c_{23}, 85s_1c_{23}, 85s_{23}]^{\mathrm{T}}$$
$$Row\ 1\text{: } (0)(85c_1c_{23}) + (0)(85s_1c_{23}) + (-c_1)(85s_{23}) \ = \ -85c_1s_{23}$$
$$Row\ 2\text{: } (0)(85c_1c_{23}) + (0)(85s_1c_{23}) + (-s_1)(85s_{23}) \ = \ -85s_1s_{23}$$
$$Row\ 3\text{: } (c_1)(85c_1c_{23}) + (s_1)(85s_1c_{23}) + (0)(85s_{23}) \ = \ 85c_{23}({c_1}^2 + {s_1}^2) \ = \ 85c_{23}$$

$$JvC_3 = \begin{bmatrix} -s_1Pc & -c_1Qc & -85c_1s_{23} \\ c_1Pc & -s_1Qc & -85s_1s_{23} \\ 0 & Pc & 85c_{23} \end{bmatrix}$$

$$J\omega C_3 = \begin{bmatrix} 0 & s_1 & s_1 \\ 0 & -c_1 & -c_1 \\ 1 & 0 & 0 \end{bmatrix}$$

Mass Matrix M(q)

The mass matrix is assembled from contributions of all three links:

$$M(q) = \Sigma_{\mathrm{i}} [\, m_{\mathrm{i}} JvC_{\mathrm{i}}^{\mathrm{T}} JvC_{\mathrm{i}} + I_{\mathrm{i}} J\omega C_{\mathrm{i}}^{\mathrm{T}} J\omega C_{\mathrm{i}} \,]$$

Link 1 Contribution

Since $JvC_1 = 0$, the translational part vanishes. Rotational part:

$$J\omega C_1^{\mathrm{T}} J\omega C_1 = [1,0,0;\ 0,0,0;\ 0,0,0]$$

$$M\_link1 = I_1 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Link 2 Contribution

Translational part $m_2\, JvC_2^T\, JvC_2$:

$$(1,1)\text{: } (-105s_1c_2)^2 + (105c_1c_2)^2 + 0 = 105^2c_2{}^2(s_1{}^2 + c_1{}^2) = 105^2c_2{}^2 = lc_2{}^2c_2{}^2$$

$$(2,2)\text{: } (-105c_1s_2)^2 + (-105s_1s_2)^2 + (105c_2)^2 = 105^2s_2{}^2 + 105^2c_2{}^2 = 105^2 = lc_2{}^2$$

$$(3,3)\text{: } 0$$

$$(1,2)\text{: } (-105s_1c_2)(-105c_1s_2) + (105c_1c_2)(-105s_1s_2) = 105^2s_1c_1c_2s_2 - 105^2c_1s_1s_2c_2 = 0$$

Rotational part $I_2\, J\omega C_2^T\, J\omega C_2$:

$$J\omega C_2^{\mathrm{T}} J\omega C_2 \text{ (1,1): } 0^2 + 0^2 + 1^2 = 1, \quad \text{(2,2): } s_1{}^2 + c_1{}^2 = 1, \quad \text{(3,3): } 0, \quad off - diag\text{: } 0$$

$$M\_link2 = m_2 \begin{bmatrix} lc_2{}^2c_2{}^2 & 0 & 0 \\ 0 & lc_2{}^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + I_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Link 3 Contribution

Translational part $m_3\, J_{vC_3}^T J_{vC_3}$:

$$(1,1)\text{: } (-s_1Pc)^2 + (c_1Pc)^2 + 0 \;=\; Pc^2$$

$$(2,2)\text{: } (-c_1Qc)^2 + (-s_1Qc)^2 + Pc^2 \;=\; Qc^2 + Pc^2 \;\; [note\text{: } Pc^2 + Qc^2 \;=\; 210^2 + 2 \cdot 210 \cdot 85c_3 + 85^2]$$

$$(3,3)\text{: } (-85c_1s_{23})^2 + (-85s_1s_{23})^2 + (85c_{23})^2 \;=\; 85^2{s_{23}}^2 + 85^2{c_{23}}^2 \;=\; 85^2 \;=\; {lc_3}^2$$

$$(2,3)\text{: } (-c_1Qc)(-85c_1s_{23}) + (-s_1Qc)(-85s_1s_{23}) + (Pc)(85c_{23})$$

$$=\; 85Qcs_{23}({c_1}^2 + {s_1}^2) + 85Pcc_{23}$$

$$=\; 85[Qcs_{23} + Pcc_{23}]$$

$$=\; 85[210(s_2s_{23} + c_2c_{23}) + 85({s_{23}}^2 + {c_{23}}^2)]$$

$$=\; 85[210cos\theta_3 \;+\; 85] \;\;=\;\; 17850c_3 \;+\; 7225 \;\;=\;\; lc_3(l_2c_3 + lc_3)$$

$$(1,2) \;=\; 0, \qquad (1,3) \;=\; 0$$

Rotational part $I_3\, J_{\omega C_3}^T J_{\omega C_3}$:

$$(1,1)\text{: } 1, \qquad (2,2)\text{: } {s_1}^2 + {c_1}^2 = 1, \qquad (3,3)\text{: } {s_1}^2 + {c_1}^2 = 1,$$

$$(2,3) = s_1s_1 + c_1c_1 = 1$$

$$M\_link3 \;\;=\;\; m_3 \begin{bmatrix} Pc^2 & 0 & 0 \\ 0 & Pc^2 + Qc^2 & 17850c_3 + 7225 \\ 0 & 17850c_3 + 7225 & 85^2 \end{bmatrix} \;+\; I_3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Final Mass Matrix M(q) = M_link1 + M_link2 + M_link3

$$M(q) \;\;=\;\; \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & M_{23} \\ 0 & M_{23} & M_{33} \end{bmatrix}$$

where:

$$M_{11} \;=\; I_1 \;+\; m_2\, {lc_2}^2{c_2}^2 \;+\; I_2 \;+\; m_3(l_2c_2 + lc_3c_{23})^2 \;+\; I_3$$

$$M_{22} \;=\; m_2\, {lc_2}^2 \;+\; I_2 \;+\; m_3({l_2}^2 \;+\; {lc_3}^2 \;+\; 2l_2lc_3c_3) \;+\; I_3$$

$$M_{33} \;=\; m_3\, {lc_3}^2 \;+\; I_3$$

$$M_{23} \;=\; M_{32} \;=\; m_3({lc_3}^2 \;+\; l_2lc_3c_3) \;+\; I_3$$

$$M_{12} \;=\; M_{13} \;=\; 0$$

Kinetic Energy K

$$K \;\;=\;\; \frac{1}{2}\, \theta^{\mathrm{T}}\, M(q)\, \theta$$

Expanding step by step. First compute M(q)θ̇:

$$M(q)\theta = [M_{11}\theta_1,\ \ M_{22}\theta_2 + M_{23}\theta_3,\ \ M_{23}\theta_2 + M_{33}\theta_3]^{\mathrm{T}}$$

Then multiply by θ^{T}:

$$K = ½[\theta_1(M_{11}\theta_1) + \theta_2(M_{22}\theta_2 + M_{23}\theta_3) + \theta_3(M_{23}\theta_2 + M_{33}\theta_3)]$$
$$= ½\, M_{11}\theta_1{}^2 + ½\, M_{22}\theta_2{}^2 + ½\, M_{23}\theta_2\theta_3 + ½\, M_{23}\theta_3\theta_2 + ½\, M_{33}\theta_3{}^2$$
$$= ½\, M_{23}\theta_2\theta_3 + ½\, M_{23}\theta_2\theta_3 = M_{23}\theta_2\theta_3\ \ [combining\ identical\ cross\ terms]$$

$$K = \frac{1}{2}M_{11}\theta_1{}^2 + \frac{1}{2}M_{22}\theta_2{}^2 + M_{23}\theta_2\theta_3 + \frac{1}{2}M_{33}\theta_3{}^2$$

Potential Energy U

Taking heights as the y (vertical) coordinate of each COM:

$$h_1 = lc_1 = 45\,mm \quad (constant)$$
$$h_2 = d_1 + lc_2\, s_2 = 90 + 105s_2$$
$$h_3 = d_1 + l_2\, s_2 + lc_3\, s_{23} = 90 + 210s_2 + 85s_{23}$$

$$U = m_1\, g\, lc_1 + m_2\, g\,(d_1 + lc_2\, s_2) + m_3\, g\,(d_1 + l_2\, s_2 + lc_3\, s_{23})$$

Lagrangian

$$L = K - U$$
$$L = ½M_{11}\theta_1{}^2 + ½M_{22}\theta_2{}^2 + M_{23}\theta_2\theta_3 + ½M_{33}\theta_3{}^2$$
$$-\, m_1 g lc_1 - m_2 g(d_1 + lc_2 s_2) - m_3 g(d_1 + l_2 s_2 + lc_3 s_{23})$$

Derivation of τ_1

Lagrange equation:

$$\tau_1 = \frac{d}{dt}\left(\frac{\partial L}{\partial \theta_1}\right) - \frac{\partial L}{\partial \theta_1}$$

Step 1: $\partial L/\partial\theta_1$

Neither K nor U contain θ_1 explicitly (M_{11}, M_{22}, M_{23}, M_{33} depend only on θ_2, θ_3; potential energy has no θ_1). Therefore:

$$\frac{\partial L}{\partial \theta_1} = 0$$

Step 2: ∂L/∂θ̇₁

$$\partial L/\partial \theta_1 \ = \ M_{11}\theta_1$$

Step 3: d/dt(M₁₁θ̇₁) — Product Rule

$$\frac{d}{dt}(M_{11}\theta_1) \ = \ \dot{M}_{11}\theta_1 \ + \ M_{11}\theta \ddot{a}_1$$

Step 4: Compute Ṁ₁₁

Recall $M_{11} = I_1 + I_2 + I_3 + m_2 lc_2^2 c_2^2 + m_3(l_2c_2+lc_3c_{23})^2$. Constants vanish. Expanding $(l_2c_2+lc_3c_{23})^2$:

$$= \ l_2^2c_2^2 \ + \ 2l_2lc_3c_2c_{23} \ + \ lc_3^2c_{23}^2$$

Variable terms: $m_2lc_2^2c_2^2$, $m_3l_2^2c_2^2$, $2m_3l_2lc_3c_2c_{23}$, $m_3lc_3^2c_{23}^2$

Differentiating each term with $\dot{c}_2=-s_2\dot{\theta}_2$ *and* $\dot{c}_{23}=-s_{23}(\dot{\theta}_2+\dot{\theta}_3)$:

$$d/dt(m_2lc_2^2c_2^2) \ = \ -2m_2lc_2^2c_2s_2\theta_2$$
$$d/dt(m_3l_2^2c_2^2) \ = \ -2m_3l_2^2c_2s_2\theta_2$$
$$d/dt(2m_3l_2lc_3c_2c_{23}) \ = \ 2m_3l_2lc_3[\dot{c}_2c_{23} + c_2\dot{c}_{23}]$$
$$= \ 2m_3l_2lc_3[(-s_2\theta_2)c_{23} \ + \ c_2(-s_{23})(\theta_2+\theta_3)]$$
$$= \ 2m_3l_2lc_3[-s_2c_{23}\theta_2 \ - \ c_2s_{23}\theta_2 \ - \ c_2s_{23}\theta_3]$$
$$= \ -2m_3l_2lc_3[(s_2c_{23}+c_2s_{23})\theta_2 \ + \ c_2s_{23}\theta_3] \ \ using\ identity\ s_2c_{23}+c_2s_{23} = s_{23}$$
$$= \ -2m_3l_2lc_3[s_{23}\theta_2 \ + \ c_2s_{23}\theta_3]$$
$$d/dt(m_3lc_3^2c_{23}^2) \ = \ m_3lc_3^2 \cdot 2c_{23} \cdot (-s_{23})(\theta_2+\theta_3) \ = \ -2m_3lc_3^2c_{23}s_{23}(\theta_2+\theta_3)$$

Collecting all terms:

$$\dot{M}_{11} \ = \ [-2m_2lc_2^2c_2s_2 \ - \ 2m_3l_2^2c_2s_2 \ - \ 2m_3l_2lc_3s_{23} \ - \ 2m_3lc_3^2c_{23}s_{23}]\theta_2$$
$$+ [-2m_3l_2lc_3c_2s_{23} \ - \ 2m_3lc_3^2c_{23}s_{23}]\theta_3$$
$$= \ -2m_3(l_2c_2+lc_3c_{23})[l_2s_2\theta_2 \ + \ lc_3s_{23}(\theta_2+\theta_3)] \ - \ 2m_2lc_2^2c_2s_2\theta_2$$

Step 5: Final τ₁

$$\tau_1 \ = \ M_{11}\theta\ddot{a}_1$$
$$-2m_2lc_2^2c_2s_2 \ \cdot \ \theta_1\theta_2$$
$$-2m_3(l_2c_2+lc_3c_{23})l_2s_2 \ \cdot \ \theta_1\theta_2$$
$$-2m_3(l_2c_2+lc_3c_{23})lc_3s_{23} \ \cdot \ \theta_1\theta_2$$
$$-2m_3(l_2c_2+lc_3c_{23})lc_3s_{23} \ \cdot \ \theta_1\theta_3$$

Derivation of τ_2

Step 1: $\partial L/\partial\dot{\theta}_2$

$$\partial L/\partial\theta_2 \;=\; M_{22}\theta_2 \;+\; M_{23}\theta_3$$

Step 2: $d/dt(M_{22}\dot{\theta}_2 + M_{23}\dot{\theta}_3)$ — Product Rule on each term

Differentiating $M_{22} = m_2lc_2^2 + I_2 + m_3(l_2^2+lc_3^2+2l_2lc_3c_3) + I_3$: only $2l_2lc_3c_3$ varies.

$$\dot{\mathrm{M}}_{22} \;=\; m_3 \cdot 2l_2lc_3 \cdot (-s_3\theta_3) \;=\; -2m_3l_2lc_3s_3\theta_3$$

$$d/dt(M_{22}\theta_2) \;=\; \dot{\mathrm{M}}_{22}\theta_2 \;+\; M_{22}\theta\ddot{a}_2 \;=\; -2m_3l_2lc_3s_3\theta_2\theta_3 \;+\; M_{22}\theta\ddot{a}_2$$

Differentiating $M_{23} = m_3(lc_3^2+l_2lc_3c_3)+I_3$: only $l_2lc_3c_3$ varies.

$$\dot{\mathrm{M}}_{23} \;=\; m_3l_2lc_3 \cdot (-s_3\theta_3) \;=\; -m_3l_2lc_3s_3\theta_3$$

$$d/dt(M_{23}\theta_3) \;=\; \dot{\mathrm{M}}_{23}\theta_3 \;+\; M_{23}\theta\ddot{a}_3 \;=\; -m_3l_2lc_3s_3{\theta_3}^2 \;+\; M_{23}\theta\ddot{a}_3$$

Combining:

$$d/dt(\partial L/\partial\theta_2) \;=\; M_{22}\theta\ddot{a}_2 \;+\; M_{23}\theta\ddot{a}_3 \;-\; 2m_3l_2lc_3s_3\theta_2\theta_3 \;-\; m_3l_2lc_3s_3{\theta_3}^2$$

Step 3: $\partial L/\partial\theta_2$

Only M_{11} depends on θ_2:

$$\partial K/\partial\theta_2 \;=\; ½(\partial M_{11}/\partial\theta_2){\theta_1}^2$$

$$\partial M_{11}/\partial\theta_2 \;=\; -2m_2{lc_2}^2c_2s_2 \;-\; 2m_3(l_2c_2 + lc_3c_{23})(l_2s_2 + lc_3s_{23})$$

$$\partial U/\partial\theta_2 \;=\; m_2glc_2c_2 \;+\; m_3g(l_2c_2 + lc_3c_{23})$$

Step 4: Final τ_2

$$\begin{aligned}\tau_2 \;\;=\;\; & M_{22}\theta\ddot{a}_2 \;+\; M_{23}\theta\ddot{a}_3 \\ & - 2m_3l_2lc_3s_3 \;\cdot\; \theta_2\theta_3 \\ & - m_3l_2lc_3s_3 \;\cdot\; {\theta_3}^2 \\ & + [m_2{lc_2}^2c_2s_2 \;+\; m_3(l_2c_2 + lc_3c_{23})(l_2s_2 + lc_3s_{23})] \;\cdot\; {\theta_1}^2 \\ & + m_2glc_2c_2 \;+\; m_3g(l_2c_2 + lc_3c_{23})\end{aligned}$$

Derivation of τ_3

Step 1: $\partial L/\partial\dot{\theta}_3$

$$\partial L/\partial\theta_3 \;=\; M_{23}\theta_2 \;+\; M_{33}\theta_3$$

Step 2: d/dt($M_{23}\dot\theta_2 + M_{33}\dot\theta_3$)

$$d/dt(M_{23}\theta_2) = \dot{\mathrm{M}}_{23}\theta_2 + M_{23}\theta\ddot{\mathrm{a}}_2 = -m_3 l_2 lc_3 s_3 \theta_2\theta_3 + M_{23}\theta\ddot{\mathrm{a}}_2$$

$$d/dt(M_{33}\theta_3) = \dot{\mathrm{M}}_{33}\theta_3 + M_{33}\theta\ddot{\mathrm{a}}_3 \; [M_{33} = m_3 lc_3{}^2 + I_3 = constant \implies \dot{\mathrm{M}}_{33} = 0] = M_{33}\theta\ddot{\mathrm{a}}_3$$

$$d/dt(\partial L/\partial\theta_3) = M_{23}\theta\ddot{\mathrm{a}}_2 + M_{33}\theta\ddot{\mathrm{a}}_3 - m_3 l_2 lc_3 s_3 \theta_2\theta_3$$

Step 3: ∂L/∂θ_3

Three elements M_{11}, M_{22}, M_{23} depend on θ_3:

$$\partial K/\partial\theta_3 = ½(\partial M_{11}/\partial\theta_3)\theta_1{}^2 + ½(\partial M_{22}/\partial\theta_3)\theta_2{}^2 + (\partial M_{23}/\partial\theta_3)\theta_2\theta_3$$

$$\partial M_{11}/\partial\theta_3 = -2m_3 lc_3(l_2c_2 + lc_3c_{23})s_{23}$$

$$\partial M_{22}/\partial\theta_3 = -2m_3 l_2 lc_3 s_3$$

$$\partial M_{23}/\partial\theta_3 = -m_3 l_2 lc_3 s_3$$

$$\partial U/\partial\theta_3 = m_3 g lc_3 c_{23}$$

$$\partial L/\partial\theta_3 = -m_3 lc_3(l_2c_2 + lc_3c_{23})s_{23}\theta_1{}^2 - m_3 l_2 lc_3 s_3\theta_2{}^2 - m_3 l_2 lc_3 s_3\theta_2\theta_3 - m_3 g lc_3 c_{23}$$

Step 4: Combine — note $\dot\theta_2\dot\theta_3$ terms cancel

From Step 2: $-m_3 l_2 lc_3 s_3 \dot\theta_2\dot\theta_3$

From Step 3 (negated): $+m_3 l_2 lc_3 s_3 \dot\theta_2\dot\theta_3$

Sum = 0 ⟹ $\dot\theta_2\dot\theta_3$ terms cancel exactly. ✓

Step 5: Final τ_3

$$\begin{aligned}\tau_3 = {} & M_{23}\theta\ddot{\mathrm{a}}_2 + M_{33}\theta\ddot{\mathrm{a}}_3 \\ & + m_3 lc_3(l_2c_2 + lc_3c_{23})s_{23} \cdot \theta_1{}^2 \\ & + m_3 l_2 lc_3 s_3 \cdot \theta_2{}^2 \\ & + m_3 g lc_3 c_{23}\end{aligned}$$

Extraction of M(q), B(q), C(q), G(q)

Velocity vectors: [q̇q] = [$\dot\theta_1\dot\theta_2$, $\dot\theta_1\dot\theta_3$, $\dot\theta_2\dot\theta_3$]^T (mixed), [q²] = [$\dot\theta_1{}^2$, $\dot\theta_2{}^2$, $\dot\theta_3{}^2$]^T (squared)

Mass Matrix M(q) — from θä coefficients:

$$M(q) = \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & M_{23} \\ 0 & M_{23} & M_{33} \end{bmatrix}$$

Coriolis Matrix B(q) — from mixed velocity products:

$$B(q) = \begin{bmatrix} -2(m_2 lc_2^2 c_2 s_2 + m_3 Pc \cdot Qs) & -2m_3 Pc \cdot lc_3 s_{23} & 0 \\ 0 & 0 & -2m_3 l_2 lc_3 s_3 \\ 0 & 0 & 0 \end{bmatrix}$$

where $Pc = l_2c_2 + lc_3c_{23}$ and $Qs = l_2s_2 + lc_3s_{23}$

Centrifugal Matrix C(q) — from squared velocity terms:

$$C(q) = \begin{bmatrix} 0 & 0 & 0 \\ (m_2 lc_2^2 c_2 s_2 + m_3 Pc \cdot Qs) & 0 & -m_3 l_2 lc_3 s_3 \\ m_3 lc_3 Pc \cdot s_{23} & m_3 l_2 lc_3 s_3 & 0 \end{bmatrix}$$

Gravity Vector G(q):

$$G(q) = \begin{bmatrix} 0 \\ m_2 g lc_2 c_2 \ + \ m_3 g(l_2 c_2 + lc_3 c_{23}) \\ m_3 g lc_3 c_{23} \end{bmatrix}$$

Note: $G_1 = 0$ because joint 1 rotates about the vertical axis and performs no work against gravity regardless of θ_1. This is physically consistent and serves as a verification checkpoint.

MATLAB Simulation and Validation:

Introduction

To verify the analytical derivations, a complete simulation model of the proposed **3DOF Articulated Manipulator** was developed using **MATLAB R2023a** and the **Robotics System Toolbox**. The simulation environment was used to validate the kinematic model, Jacobian matrix, workspace, singularity analysis, and dynamic model before implementing the manipulator on physical hardware.

The simulation followed the same Denavit–Hartenberg (DH) parameters, physical dimensions, and mass properties used in the analytical derivations to ensure consistency between theory and implementation.

Robot Model Development

The first step involved creating a rigid body tree model of the manipulator using MATLAB. Three revolute joints were defined according to the standard DH convention.

The manipulator dimensions used in the simulation are:

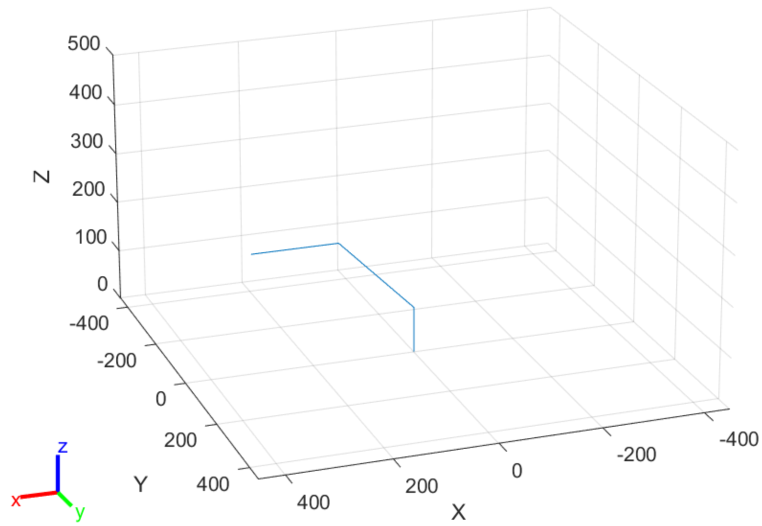
- Base height = **90 mm**
- Link 2 length = **210 mm**

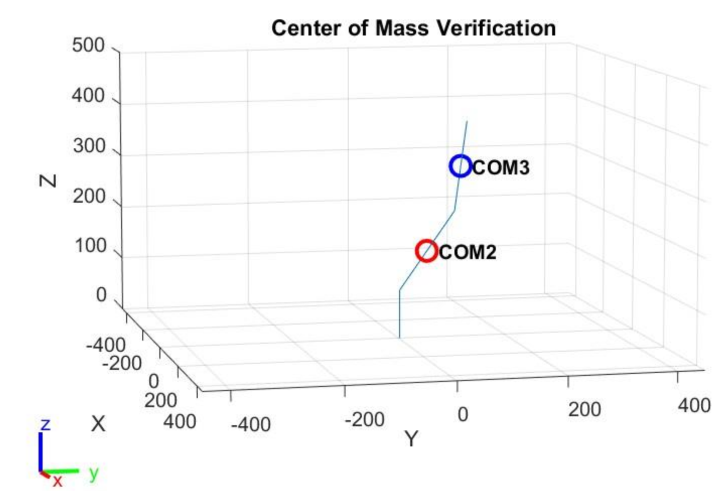
- Link 3 length = **170 mm**

The home configuration selected for the project is

$$[0°, 45°, -45°]$$

This configuration was intentionally selected because the original configuration $[0°, 0°, 0°]$was found to be singular during Jacobian analysis.





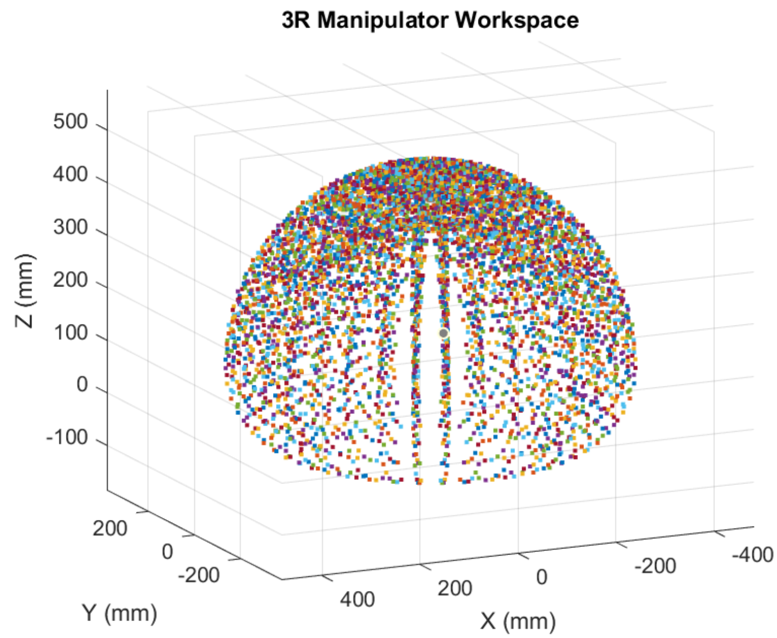
Workspace Analysis

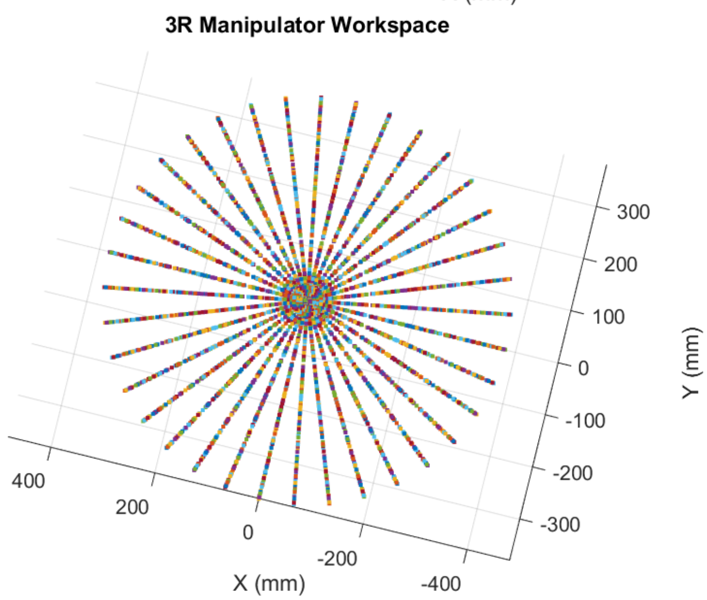
The workspace of the manipulator was generated by sweeping all three joints within their allowable ranges.

Joint limits used during simulation were:

- Joint 1: 0° to 180°
- Joint 2: 0° to 180°
- Joint 3: –90° to +90°

The generated workspace illustrates all reachable end-effector positions.

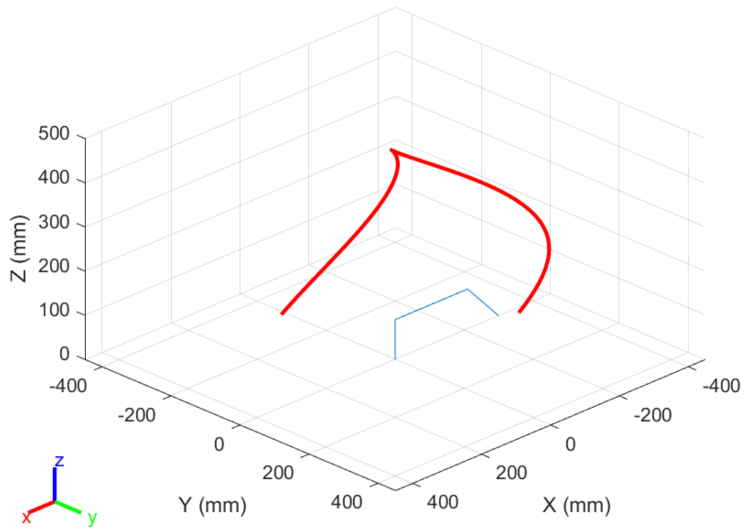




End-Effector Trajectory

A smooth trajectory was generated to observe the continuous movement of the end effector.

The trajectory remained continuous throughout the simulation without discontinuities, confirming that the analytical forward kinematics produced physically consistent motion.



Workspace Manipulability Analysis

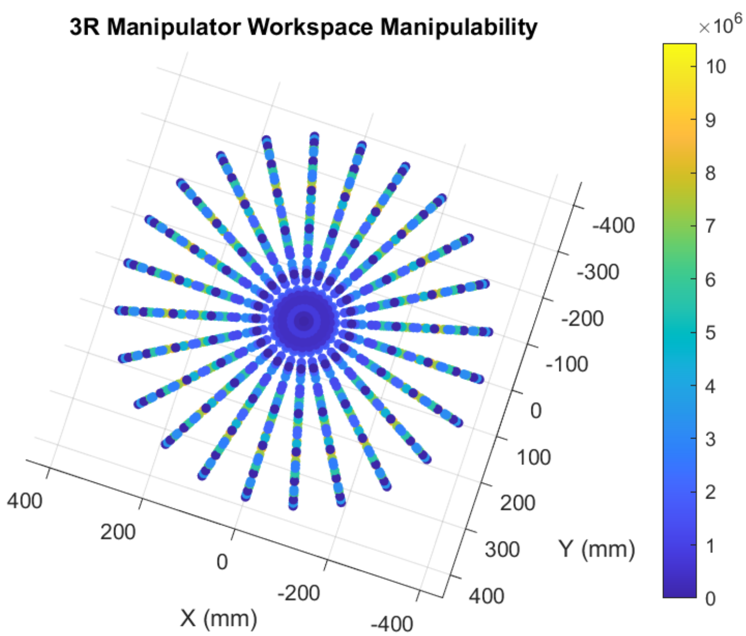
To evaluate dexterity throughout the workspace, the manipulability index

$$w = \sqrt{\det(J_v J_v^T)}$$

was calculated for a large number of reachable configurations.

A color-coded workspace map was generated in MATLAB, where larger values indicate higher dexterity and smaller values identify regions approaching singularities.

This visualization assists in selecting safe operating regions for trajectory planning and practical operation.



Dynamic Model Verification

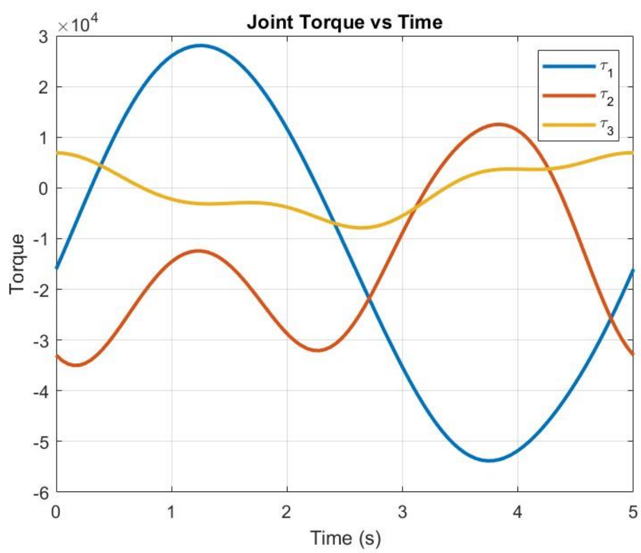
The dynamic parameters obtained analytically were implemented in MATLAB.

The following quantities were evaluated:

- Mass matrix $M(q)$
- Gravity vector $G(q)$
- Joint torque equations
- Symmetry of the inertia matrix
- Positive definiteness of the inertia matrix

The simulation verified that:

- The mass matrix is symmetric.
- All eigenvalues are positive.
- The gravity vector agrees with the analytical model.
- The computed joint torques match the derived equations of motion.



Control:

Basic Revision

PID Controller

A Proportional-Integral-Derivative (PID) controller produces a control output u(t) from the joint angle error e(t) = θ_d(t) − θ(t):

$$u(t) \; = \; K_p e(t) \; + \; K_i \int e(t)dt \; + \; K_d \frac{de}{dt}$$

In discrete form (implemented on the ESP32 at sampling period T_s):

$$u[k] = K_p e[k] + K_i T_s \Sigma e[k] + K_d \frac{e[k] - e[k-1]}{T_s}$$

State-Space Representation

Single-joint dynamics in state-space form with state x = [θ, θ]^T:

$$\dot{x} = Ax + Bu, \quad y = Cx$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -b/J \end{bmatrix}, \quad B = [\ 01/J], \quad C = [1 \quad 0]$$

where J is the effective moment of inertia at the joint and b is the viscous friction coefficient.

Robot Control Strategies

Joint-Space Control

The simplest strategy treats each joint independently with a PID loop. The ESP32 maps target angle to PWM pulse width linearly:

$$Pulse\ width\ (\mu s) = 500 + \frac{\theta_i}{180°} \times 2000$$

This approach ignores dynamic coupling between joints and is acceptable for slow, precise motions under light loading.

Computed Torque Method

The full dynamic model is inverted to linearise and decouple the system:

$$\tau = M(q)(\ddot{a}q\_d \quad + K_d\dot{e} + K_p e) + C(q,\dot{q})q + G(q)$$

This cancels the nonlinear dynamics, reducing the closed-loop system to three decoupled linear second-order systems. Accurate values of M, C, G (requiring measured link masses) are needed.

Task-Space Control

The Cartesian position error drives the control via the Jacobian inverse:

$$\dot{q} = J^{v-1}\left(\dot{x}\_d + K_p(x\_d - x_e)\right)$$

The pseudo-inverse $J_v^+ = J_v^T(J_vJ_v^T)^{-1}$ is used near singular configurations. This strategy is preferred for straight-line trajectory tracking in Cartesian space.

Aurduino Implementation Notes

- PWM frame frequency: 50 Hz on all three servo channels.
- Pulse width range: 500 μs (0°) to 2500 μs (180°).
- Control loop runs on Core 1; serial communication and trajectory interpolation run on Core 0.
- The ledc peripheral provides 14-bit PWM resolution (≈0.01° per step).
- On power-up, all joints are driven to home position before accepting commands.
- Software joint-limit enforcement prevents commands outside the 0°–180° range.

9. Social and Economic Issues

9.1 Employment and Workforce Displacement

The increasing deployment of robotic manipulators in manufacturing and logistics raises concerns about workforce displacement. Repetitive assembly, welding, and pick-and-place tasks are most susceptible to automation. While robots reduce unit production cost and improve repeatability, they can displace semi-skilled workers in the short term. The economic consensus favours net long-term job creation through new industries such as robot manufacturing, programming, and maintenance, but the transition period requires retraining programmes and active policy support.

9.2 Safety

Robotic manipulators pose physical hazards including collision, pinch-point injuries, and unpredictable motion during faults. International standards (ISO 10218, ISO/TS 15066) mandate safety fencing, emergency stops, and force-torque limiting for collaborative robots. The 3R arm presented here operates at low torque (MG995 maximum 9.4 kg·cm) and low speed, placing it in the low-risk category. Safe operating zones, software joint-limit enforcement, and power-off-on-fault logic are implemented in the ESP32 firmware.

9.3 Economic Viability and Cost

A principal advantage of this design is its low cost relative to commercial 3-DOF arms. The bill of materials — three MG995 servos, acrylic sheet, ESP32 board, fasteners, and a power supply — totals approximately USD 15–25, compared with thousands of dollars for a commercial counterpart. This makes the platform accessible to university laboratories, STEM education programmes, and small-scale research groups in developing economies, democratising access to robotic manipulation technology.

9.4 Educational and Research Value

The project provides a tangible, hands-on platform for teaching kinematic theory (DH parameters, FK, IK), velocity analysis (Jacobian, singularities), and basic dynamics — concepts that are otherwise purely abstract in textbook form. Students engage directly with the gap between theoretical models and physical hardware constraints such as servo backlash, acrylic flex, and PWM resolution limitations.

9.5 Environmental Impact

Acrylic (PMMA) is a petrochemical-derived thermoplastic that is not readily biodegradable. Responsible disposal, recycling through specialist PMMA processors, or substitution with bio-based polymers should be considered at end-of-life. The low power consumption of the ESP32 and MG995 servos (total peak current below 3 A at 6 V, i.e. 18 W maximum) results in a minimal operational carbon footprint, especially when powered from renewable energy sources.

9.6 Intellectual Property and Open-Source Ethics

The design, firmware, and kinematic analysis developed in this project are intended for open academic dissemination. Sharing CAD files, MATLAB scripts, and Arduino code under an open-source licence (MIT or Creative Commons) promotes reproducibility and allows other researchers to build upon the work without intellectual property barriers — an important principle in publicly funded academic research.

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